

**Be able to recognize AND use the following formulas:**

- The average value of a function  $f$  on the interval  $[a, b]$  is given by  $\frac{1}{b-a} \int_a^b f(x) dx$
- The *length of the arc* on the graph of  $y = f(x)$  from the point  $(a, f(a))$  to the point  $(b, f(b))$  is  $\pi \int_a^b \sqrt{1 + [f'(x)]^2} dx$
- The area of a region between  $f(x)$  and  $g(x)$  on  $[a, b]$  is  $\int_a^b (f(x) - g(x)) dx$
- The volume of the solid generated by revolving the region bound by the  $x$ -axis and  $f(x)$  from  $[a, b]$  is  $\pi \int_a^b (f(x))^2 dx$ .
- Integration by parts  $\int u dv = uv - \int v du$ .

**Be able to recognize and use the following properties of integration:**

- FTC part 1:  $F(x) = \int_a^x f(t) dt$ , this shows that an indefinite integral can be reversed by differentiation (and vice-versa up to a constant).  
ex. If  $F(x) = \int_0^x t^3 dt$  then  $\frac{d}{dx} F(x) = \int_0^x t^3 dt = x^3$ , and If  $F(x) = \int_0^{x^2} t^3 dt$  then  $\frac{d}{dx} F(x^2) = \int_0^{x^2} t^3 dt = 2x(x^2)^3 = 2x^7$
- FTC part 2:  $\int_a^b f(x) dx = F(b) - F(a)$  where  $\frac{d}{dx} F(x) = f(x)$  on  $[a, b]$ .  
ex.  $\int_2^5 x^2 dx = \frac{x^3}{3} \Big|_2^5 = \frac{5^3}{3} - \frac{2^3}{3} = 39$
- If  $\int_a^b f(x) dx = A$  then  $\int_b^a f(x) dx = -A$
- $\int_a^b f(x) + g(x) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$ .
- $\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$  if  $a \leq b \leq c$ .
- If  $f(x)$  is an *even* function, i.e.,  $f(x) = f(-x)$ , then  $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ .
- If  $f(x)$  is an *odd* function, i.e.,  $f(x) = -f(-x)$ , then  $\int_{-a}^a f(x) dx = 0$ .

**Be able to recognize the techniques needed to integrate and be able to use them:**

- substitution.
- integration by parts
- partial fractions
- trigonometric powers
- the trigonometric substitution using  $\sqrt{a^2 - x^2} = a \cos \theta$ .

**Be able to complete the following type of problems.**

- Initial value problems  
ex.  $\frac{dy}{dx} = (x+2)^{3/2}$  and  $y(0) = 1$ .
- Definite integral of an absolute value.
- Definite integral from a graph.

**Exam 1**

Evaluate the following using the provided values.

$$\int_0^{\pi/2} g(x) dx = 1, \int_{\pi/2}^{\pi} g(x) dx = \pi - 1, \int_{\pi}^{3\pi/2} g(x) dx = \pi + 1, \int_{3\pi/2}^{2\pi} g(x) dx = 2\pi - 1$$

1.  $\int_0^{\pi} g(x) dx$
2.  $\int_{\pi}^0 \frac{g(x)}{\pi} dx$
3. Suppose that  $g(x)$  is *even*,  
find  $\int_{-\pi/2}^{\pi/2} g(x) dx$ .

Use geometry or symmetry to evaluate the definite integral. Recall that the area of a triangle is  $A = \frac{1}{2}bh$ .

4.  $\int_{-2}^2 (2 - |x|) dx$

Evaluate the following.

5. Let  $a$  and  $b$  be constants.

- a.  $\frac{d}{dx} \int_a^x 3t^2 dt$

- b.  $\frac{d}{dx} \int_a^b f(t) dt$

6.  $\int_0^2 (x^2 + 1) dx$
7.  $\int_0^{\pi} \frac{1}{2} \sin \frac{\theta}{2} d\theta$ .
8.  $\int \frac{1 - z^4}{z^2} dz$
9.  $\int x^3 (x^4 + 4)^{15} dx$
10.  $\int \frac{1}{(5x-3)^2} dx$
11.  $\int \cos x \sin^3 x dx$

Recall that  $\sin x$  is an odd function.

12.  $\int_{-5}^5 \sin^3 x dx$

Solve the following word problems.

13. An arch is 10 ft high and a 20 ft base. Its shape can be modeled by  $y = 10 - \frac{x^2}{10}$ . Find the average height of the arch above the ground. Note that the maximum height is at  $x = 0$ .

**Exam 2**

Find the displacement over the given interval.

1.  $v(t) = \cos\left(\frac{\pi}{2}t\right)$  on  $[0, 3]$ ;  $s(0) = 0$ .

Determine the position function.

2. An object is moving at a velocity of  $(t + 9)^{1/2}$  with an initial position of  $s(0) = 20$  at time  $t$ .

Solve the problem.

3. A projectile is launched vertically from the ground at  $t = 0$  where  $t$  is seconds, and its velocity in flight (in m/s) is given by  $v(t) = 20 - 10t$ .
  - a. Find the position at time  $t$  for  $0 \leq t \leq 4$ .
  - b. Find the total distance traveled from  $t = 0$  to  $t = 4$ .

Use calculus to find the area of the region described.

4. The region bounded by  $y = x$  and  $y = x^2 - 4x$ .

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5. The region bounded by  $y = \frac{1}{2}x^2$  and  $y = -x^2 + 6$ .

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6. The region in the first quadrant bounded by  $-x + 1$  and  $x - 1$  from  $0 \leq x \leq 2$ .

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Let  $R$  be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when  $R$  is revolved about the  $x$ -axis.

7.  $y = \sqrt{x}$ ,  $x = 0$ , and  $x = 9$ .

8.  $y = \sqrt{\sin 4x}$ ,  $y = 0$ ,  $0 \leq x \leq \frac{\pi}{4}$ .

Find the length of the curve.

9.  $y = 2x^{3/2}$  from  $x = 0$  to  $x = \frac{5}{4}$ .

10.  $y = \left(4 - x^{2/3}\right)^{3/2}$  from  $x = 0$  to  $x = 8$ .

### Exam 3

Perform the indicated operation.

11.  $\int \left(10e^{5x} + \frac{1}{x}\right) dx$

12.  $\int \frac{dx}{3x - 5}$

Perform the indicated operation.

13.  $\int 2xe^{2x} dx$

14.  $\int x \cos(8x) dx$

15.  $\int 10x \ln(2x) dx$

### Exam 4

1.  $\int \cos^3 2x dx$

2.  $\int \sin^2 x dx$

3.  $\int \sin^3 x \cos x dx$

4.  $\int \frac{dx}{\sqrt{25 - x^2}}$

5.  $\int_0^7 \sqrt{49 - x^2} dx$ . (Hint:  $\int \cos^2 x dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$ )

6.  $\int \frac{x}{x^2 + x - 12} dx$

7.  $\int_0^{\ln 2} 4xe^x dx$